A second challenge is building a strong domestic market for locally designed chips. Greater adoption across sectors such as transportation, consumer electronics, aerospace, and security can help companies establish sustainable product lines while remaining globally competitive. Finally, while India's semiconductor manufacturing ecosystem is growing, companies developing advanced chips still rely heavily on global foundries, making international manufacturing partnerships critical for moving designs from concept to silicon production.

EFY: What challenges do startups in India face when bringing silicon products from concept to tape-out?

SS: The first challenge is securing a strong go-to-market strategy, adequate funding, and ideally an anchor customer who believes in the product idea, as many startups struggle to cross this initial business hurdle. Once that is addressed, finding the right talent becomes critical. Chip development requires engineers with a system-level understanding who can visualise what the chip needs to achieve electronically and translate those goals into silicon, rather than simply executing individual tasks within an engineering project. The third major challenge is managing risks related to time-to-market, tape-out schedules, and execution. Tape-out is not the finish line—it's where the real validation begins.

On fabs and nodes

EFY: What technical challenges must new fabs overcome before reaching stable high-volume production?

SS: One of the biggest challenges for a new fab is achieving high yield, which is a key measure of manufacturing success. Yield is the percentage of chips on a wafer that function correctly, and reaching 98-99 per cent requires multiple rounds of prototyping, testing, and optimisation. From a design perspective, the process begins with transistor development, followed by IP creation and test chip fabrication. These test chips are then manufactured and tested, and the resulting post-silicon data is used to refine the design and production flow.

At the same time, a fab depends on a much larger ecosystem involving equipment, materials, gases, chemicals, and manufacturing processes, all of which must work together reliably. Because of this complexity, bringing a fab to stable high-volume production is a long and systematic effort. It requires continuous tuning across multiple stages and cannot be achieved quickly just by setting up a manufacturing facility.

> **ENGINEERS MUST GO BEYOND THE TRADITIONAL FOCUS ON AREA, POWER, AND COST SEEN IN MATURE NODES AND ANALYSE MANY MORE INTERACTING FACTORS SUCH AS POWER, PERFORMANCE, THERMAL BEHAVIOUR, NOISE, STRESS, AND COOLING.**

EFY: How is the market split across technology nodes, and what does that mean for design?

SS: Looking at technology nodes, a large share of chips in the market is still built on mature nodes. Around 60 to 65 per cent of chips use mature nodes, which here means 12 nm up to 28 nm (for example, 12, 16, 20 and 28 nm). Only about 35 per cent of the market is at the cutting edge; within that, truly advanced nodes—around 2 nm and below—account for just 2 to 3 per cent. These leading-edge designs are mainly used in high-end data centre and mobile applications, where advanced performance is required. These figures are consistent with public industry data, and they show that semiconductor design continues to evolve across both mature and cutting-edge nodes rather than shifting entirely to the newest technology.

On below 3 nm and beyond

EFY: What new design challenges emerge when moving below 3 nm?

SS: Moving below 3 nm into angstrom-based design shifts the challenges considerably. Higher transistor density packs more devices into the same area, raising power density and intensifying thermal issues and mechanical stress inside the chip. Power delivery becomes more complex, especially in how power grids are designed to support fast transistor switching without instability.

Engineers must therefore go beyond the traditional focus on area, power, and cost seen in mature nodes and analyse many more interacting factors such as power, performance, thermal behaviour, noise, stress, and cooling. This also requires more advanced methods like dynamic voltage and thermal analysis, and even cooling must be modelled as part of the chip design itself. In short, design below 3 nm adds multiple tightly coupled variables that need deep, simultaneous optimisation.

EFY: What EDA innovations do gate-all-around (GAA) transistor architectures need, and which technology areas are driving EDA demand in India and globally?

SS: Electronic design automation (EDA) innovation for GAA architectures is mainly focused on how libraries are designed and implemented. IP teams develop embedded memory and logic libraries for different foundries across technologies